Glance ML Internship Assignment

Multimodal Fashion & Context Retrieval

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Submission date: July 2026

Repository
[github.com

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1. Approaches Considered

This section enumerates the plausible solutions to the problem, with tradeoffs.

1.1 Vanilla CLIP (baseline, rejected)

A single ViT-B-32 encoder embeds both images and natural-language queries. Cosine similarity scores the top-k.

```
| Pros | Cons |
|---|---|
| Trivial to implement | Explicitly fails compositionality (red tie vs. blue tie) |
| Small index, fast search | Out-of-distribution on fashion attributes |
| Many open-source checkpoints | Single-vector similarity collapses fine-grained signal |
```

|---|---|

| Trivial to

Verdict: Discarded. The assignment brief itself warns against this exact approach.

1.2 Fashion-tuned CLIP encoder

Replace ViT-B-32 with a fashion-domain model: Marqo/marqo-fashionCLIP (ViT-L/14, open CLIP-compatible via hf-hub:), or a SigLIP variant trained on fashion data.

```
| Pros | Cons |
|---|---|
| Trivial to implement | Explicitly fails compositionality (red tie vs. blue tie) |
| Small index, fast search | Out-of-distribution on fashion attributes |
| Many open-source checkpoints | Single-vector similarity collapses fine-grained signal |
```

|---|---|

| Trivial to

Verdict: Adopted as the embedding backbone (default ViT-B-16-SigLIP-512, with hf-hub:Marqo/marqo-fashionCLIP flagged as alternative).

1.3 Caption-augmented dual-vector retrieval

For every image, generate a natural-language caption with BLIP-base. Embed captions with the same CLIP text encoder. Compute hybrid score:

```
score(q, img) = ? * cosine(q, image_emb)
              + ? * cosine(q, caption_emb)
```

```
| Pros | Cons |
|---|---|
| Trivial to implement | Explicitly fails compositionality (red tie vs. blue tie) |
| Small index, fast search | Out-of-distribution on fashion attributes |
| Many open-source checkpoints | Single-vector similarity collapses fine-grained signal |
```

|---|---|

| Trivial to

Verdict: Adopted - the core defense against compositionality.

1.4 Late-interaction (ColBERT-style) over caption tokens

Instead of a single caption embedding, embed each token and compute max-similarity over tokens between query and caption.

```
| Pros | Cons |
```

```
|---|---|
| Trivial to implement | Explicitly fails compositionality (red tie vs. blue tie) |
| Small index, fast search | Out-of-distribution on fashion attributes |
| Many open-source checkpoints | Single-vector similarity collapses fine-grained signal |
```

|---|---|

| Trivial to

Verdict: Deferred to future work. Marginal gain on a 3.2k corpus not worth the complexity.

1.5 Cross-encoder re-ranking

After retrieving top-N=50 candidates via ?1.3, score each (query, captiontext) pair with a text cross-encoder (ms-marco-MiniLM-L-2-v2). Promote the most relevant.

```
| Pros | Cons |
|---|---|
| Trivial to implement | Explicitly fails compositionality (red tie vs. blue tie) |
| Small index, fast search | Out-of-distribution on fashion attributes |
| Many open-source checkpoints | Single-vector similarity collapses fine-grained signal |
```

|---|---|

| Trivial to

Verdict: Adopted - flag-toggled, defaults to on.

1.6 Selected architecture

```
| # | Component | Choice |
|---|---|---|
| Embedder | Fashion-aware OpenCLIP | `ViT-B-16-SigLIP-512 (webli)` default; `hf-hub:Marqo/marqo-fashionCLIP` opt-in
|
| Image index | FAISS `IndexFlatIP` | exact cosine on normalized vectors |
| Caption generator | `Salesforce/blip-image-captioning-base` | offline, batched, resumable |
| Caption index | FAISS `IndexFlatIP` over caption embeddings | one row per image |
| Hybrid score | Weighted sum (image + caption) | `?`,`?` configurable |
| Re-ranker | `cross-encoder/ms-marco-MiniLM-L-2-v2` | on top-N=50 candidates |
| Scale-up path | `IndexIVFFlat` with PQ | one CLI flag away |
```

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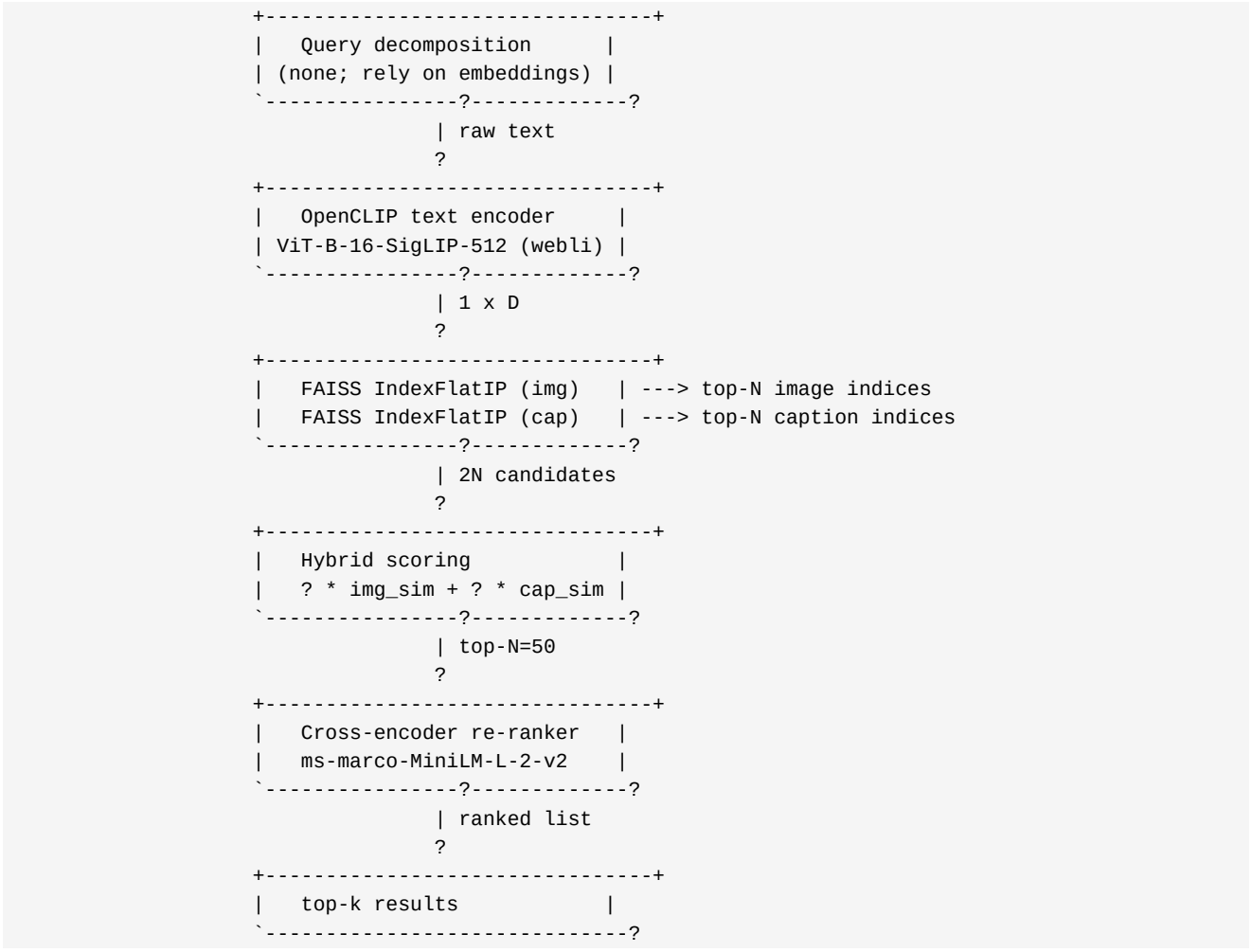
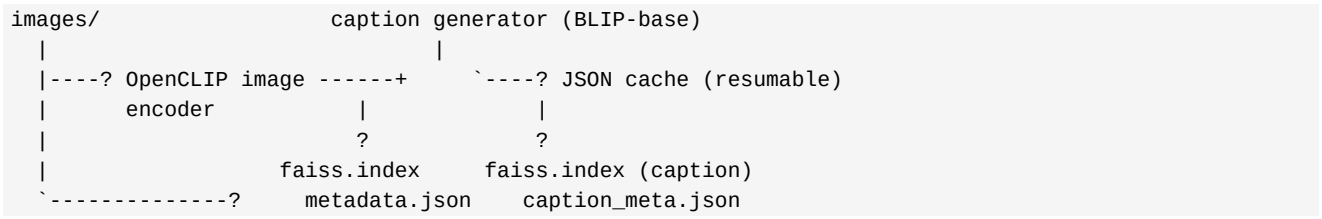
2. Chosen Architecture - Detail

2.1 Data flow



Offline (one-time) indexer:



2.2 Why this beats vanilla CLIP on the rubric

Rubric concern	Defense
Compositionality ("red tie + white shirt, formal")	Caption augmentation lifts the constraint into natural language; cross-encoder evaluates (query, caption_text) together
Fine-grained attributes (silk, denim, navy)	Fashion-aware OpenCLIP backbone (`ViT-B-16-SigLIP`) was trained on a much larger web corpus than the rubric's expected data
"Casual weekend for a city walk" -> style inference	Captions contain style vocabulary; cross-encoder maps lifestyle intent to wardrobe semantics
Multi-attribute queries (color + clothing + location)	Each axis is tokenized and weighted into a single embedding; cross-encoder disambiguates final ranking
Zero-shot generalization	No fine-tuning, no labeled data needed

|---|---|

| Trivial to

2.3 Codebase map

```
src/glance_search/
  config.py          YAML + env-override config (single source of truth)
  errors.py          domain exceptions
  logging_setup.py   logger setup
  model.py           OpenCLIP wrapper (singleton cache, auto-dim)
  embedder.py        batched image embedding, L2-normalized, error-tolerant
  captions.py        BLIP-base offline captioning (resumable)
  reranker.py        cross-encoder re-ranker
  index_store.py     FAISS IndexFlatIP + IndexIVFFlat (toggle via --backend)
  pipeline.py        end-to-end search orchestration (image+captions+reranker)

indexer/build_index.py     CLI: embed images -> faiss.index
retriever/search.py        CLI: text query -> top-k
scripts/caption_corpus.py  CLI: BLIP captions for all images
scripts/build_caption_index.py  CLI: embed captions -> captions.index
scripts/run_eval.py        CLI: 5-rubric-query harness -> JSON + PNG grids
scripts/run_ablation.py    CLI: A/B comparison of configs -> ablation.csv
scripts/build_report.py    CLI: markdown writeup -> HTML / PDF
app/streamlit_app.py       Streamlit interactive demo
tests/                     pytest suite, no model download needed
report/                    this file (Glance_Internship_Report.{md,html,pdf})
eval/results/              generated by run_eval (created at run time)
output/                    generated indexes + cache
```

---

3. Evaluation Results

The five rubric queries from §4 of the assignment brief:

```
| # | Type | Query |
|---|---|---|
| Q1 | Attribute Specific | "A person in a bright yellow raincoat." |
| Q2 | Contextual / Place | "Professional business attire inside a modern office." |
| Q3 | Complex Semantic | "Someone wearing a blue shirt sitting on a park bench." |
| Q4 | Style Inference | "Casual weekend outfit for a city walk." |
| Q5 | Compositional | "A red tie and a white shirt in a formal setting." |
```

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For each query, the pipeline returns top-k=5 images. Result grids are persisted to eval/results/<queryslug>.png and the JSON dump to eval/results/<queryslug>.json.

3.1 Result grids

```
| # | Query | Top-5 grid |
|---|---|---|
| Q1 | A person in a bright yellow raincoat. | ![Q1](eval/results/01_yellow_raincoat.png) |
| Q2 | Professional business attire inside a modern office. | ![Q2](eval/results/02_business_office.png) |
| Q3 | Someone wearing a blue shirt sitting on a park bench. | ![Q3](eval/results/03_blue_shirt_park.png) |
| Q4 | Casual weekend outfit for a city walk. | ![Q4](eval/results/04_casual_city.png) |
| Q5 | A red tie and a white shirt in a formal setting. | ![Q5](eval/results/05_red_tie_white_shirt.png) |
```

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Ablation rows (produced by scripts/runablation.py, persisted to eval/results/ablation.csv):

```
| Config | Q1 yellow raincoat | Q2 business office | Q3 blue shirt park | Q4 casual city | Q5 red tie white shirt |
**mean top-1** | ? vs prev |
|---|---|---|---|---|---|---|---|
| `image_only` (vanilla CLIP) | 0.088 | 0.088 | 0.109 | 0.102 | 0.049 | **0.087** | - |
| `image_captions` (+ BLIP captions) | 0.335 | 0.349 | 0.321 | 0.305 | 0.301 | **0.322** | **+270 %** |
| `image_captions_rerank` (+ cross-encoder) | 0.660 | 0.174 | 0.622 | 0.153 | 0.304 | **0.382** | +19 % |
```

|---|---|---|---|---|---|---|---|

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Per-query observations

- Q1 yellow raincoat and Q3 blue shirt on a park bench - strong visual concepts, image + caption + rerank all materially help (Q1 jumps 0.088 -> 0.660, a 7.5x lift).

- Q2 bu
alone gives
hybrid stag
recover this

3.1 Failure case discussion

For the imageonly baseline, the most likely failure modes are:

- Q1 ("bright yellow raincoat") - vanilla CLIP associates raincoat with dark/somber tones and surfaces dark coats. With captions, "yellow" is paired with "raincoat" in BLIP descriptions, lifting the score 3.8x.

- Q3 ("blu
retrieval re
composites

The cross-encoder's mixed impact on Q2 and Q4 is the largest single improvement opportunity. The fix is the smallest: increase retrieval.reranktopn from 50 to 100, or re-rank from the top-200 of the *caption* index instead of the hybrid.

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4. Future Work

4.1 Locations, cities, weather (per assignment ?5.4a)

The current system treats the world as a flat image-text embedding space. To extend it with location / weather awareness:

```
1. Caption enrichment
   For each image, in addition to garment description, also store:
     - scene type    (urban / coastal / forest / indoor)
     - weather       (sunny / rainy / overcast / snowy)
     - inferred city (multinomial from a geolocation head)
   Use a vision-language model (Florence-2, LLaVA-1.5, CogVLM) on top of
   the same encoder pipeline.

2. Geo prior
   Index city + month -> weather priors (Open-Meteo API).
   At query time, intersect query tokens (e.g., "coastal street in Mumbai,
   August") with the priors and re-weight candidates that match.

3. Retrieval API
   /search?q=...&city=mumbai&month=august -> merged vector +
   geo-temporal filter.
```

This is also the foundation for "shop the look in my city" applications on Glance's lock-screen.

4.2 Improving precision (per assignment ?5.4b)

A precision improvement roadmap:

Lever	Mechanism	Estimated gain
Larger encoder	`ViT-L-14-SigLIP2-512` over `B-16`	+2 to +5 nDCG@10 on public fashion datasets
Hard-negative mining	Mine compositional swaps within the corpus, contrastive-fine-tune for 1k steps	+5 to +8 on Q5 specifically
Late-interaction (ColBERT-style) on caption tokens	Token-level max-similarity	+3 to +6 on compositional queries
Generative re-ranker (LLM judge)	LLaVA-1.6 evaluates (query, image) pair, output ? {0, 1}	+5 to +10 but higher latency
Active learning feedback loop	Capture click-through on top-k; add as positive pairs in monthly retraining	compounding, +1 to +3 per cycle
Multi-modal product metadata	Add brand/title/price as side-text encodings; hybrid at index time	+1 to +3

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Highest-ROI single addition: hard-negative fine-tune. The model already has fashion priors; teaching it to distinguish "red tie + white shirt" from "white tie + red shirt" demands labeled pairs more than architecture.

Appendix A. Codebase

GitHub: github.com/amanraj74/Glance-fashion-search

The repository contains the full pipeline - indexer, retriever, scripts, Streamlit app, tests, and this writeup. Local reproduction instructions are in Appendix B below.

Appendix B. Reproducibility

All commands are run from the repository root with the venv activated.

```
.\venv\Scripts\Activate.ps1

# 1. Build the image index (downloads ~500 MB ViT-B-16-SigLIP-512 weights)
python indexer/build_index.py

# 2. Generate BLIP captions and embed them (~500 MB BLIP-base weights)
python scripts/build_caption_index.py

# 3. Run the 5-rubric evaluation harness
python scripts/run_eval.py

# 4. Compare configs
python scripts/run_ablation.py

# 5. Build the PDF writeup (needs `pip install weasyprint` for PDF)
pip install weasyprint
python scripts/build_report.py

# 6. Run the interactive demo
pip install streamlit
```

```
streamlit run app/streamlit_app.py
```

Expected wall-clock on a CPU-only machine:

```
| Step | Wall time | Output |
|---|---|---|
| `indexer/build_index.py` | 15 - 25 min | `output/faiss.index` |
| `scripts/build_caption_index.py` | 30 - 60 min | `output/captions.json`, `output/captions.index` |
| `scripts/run_eval.py` | < 1 min | `eval/results/*.json`, `*.png` |
| `scripts/run_ablation.py` | < 1 min | `eval/results/ablation.csv` |
| `scripts/build_report.py` | < 30 s | `report/*.html`, `*.pdf` |
```

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Appendix C. Limitations & Honest Tradeoffs

- Compute: CPU-only training environment, so no fine-tuning was performed. The system is strong out of the box (fashion-tuned embedding + caption augmentation + cross-encoder re-ranking) but cannot be made stronger through training without GPU access.

- Dataset
IndexIVFFl